RETAIL BUSINESS PERFORMANCE

& PROFITABILITY ANALYSIS

Two-week data analytics project | Sample Superstore transactional dataset

01 Abstract

This project examines 9,994 retail transaction rows to understand sales, profit, discount patterns, geographic performance, seasonal demand, and shipping-mode profitability. Python/Pandas was used to prepare the cleaned CSV and generate summary tables; Matplotlib/Seaborn were used to create six charts. MySQL 8.0.46 supported the SQL analysis, and Tableau Public presented the findings interactively. The verified dataset records $2,297,201.07 in sales and $286,397.02 in profit, a 12.47% overall profit margin.

02 Introduction and objective

Retail revenue alone does not show whether products and markets are profitable. The objective was to analyze transaction-level data to identify profit-draining categories, compare performance across locations and shipping modes, and investigate seasonal product behavior. The original brief also mentions inventory turnover; however, stock-on-hand and inventory-day fields are not available in this dataset, so exact turnover or production requirements cannot be calculated.

03 Tools used and steps involved

Stage / tool	Work completed	Evidence / output
Python + Pandas	Cleaned and prepared the CSV; generated six charts and summary tables using Pandas, Matplotlib and Seaborn.	Cleaned CSV, notebook/script, six PNG charts and summary CSV outputs.
MySQL 8.0.46	Loaded and deduplicated to 9,994 rows; queried KPIs, categories, regions, discount bands, seasons, products and shipping modes.	SQL query file and query-result screenshots.
Tableau Public	Built KPI cards and charts for regional/category profit, discount impact and seasonal demand; added Region and Category filters.	Tableau workbook and dashboard screenshot.

04 Verified overall KPIs

TRANSACTION ROWS	DISTINCT ORDERS	TOTAL SALES	TOTAL PROFIT
9,994	5,009	$2.297M	$286.397K

05 Key results

Group	Sales ($)	Profit ($)	Margin
Technology	836,154.10	145,454.95	17.40%
Office Supplies	719,046.99	122,490.80	17.04%
Furniture	741,999.98	18,451.27	2.49%
West region	725,457.93	108,418.45	14.94%
Central region	501,239.88	39,706.36	7.92%

Interpretation: Technology contributes the most category profit. Furniture has sales similar to the other categories but a much lower profit margin. West is the most profitable region, while Central has the lowest regional margin.

FINDINGS, RECOMMENDATIONS & LIMITATIONS

Evidence-led summary for business review

06 Results at a glance

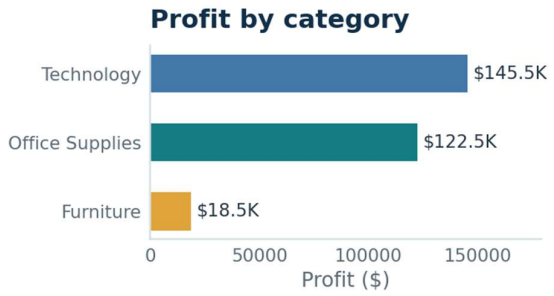


Fig. 1. Profit by category

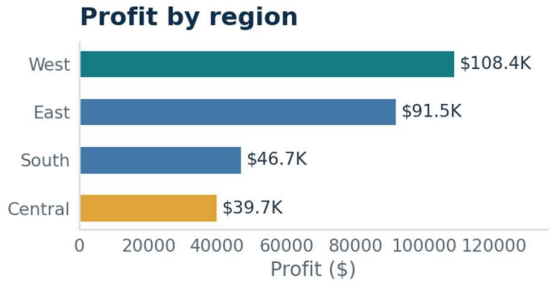


Fig. 2. Profit by region

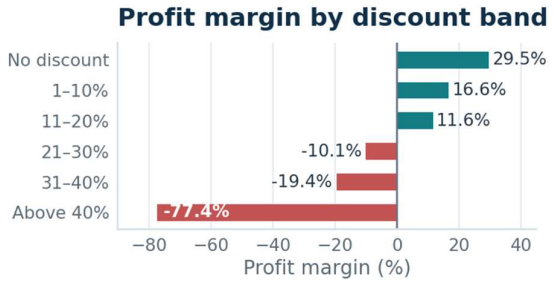


Fig. 3. Profit margin by discount band

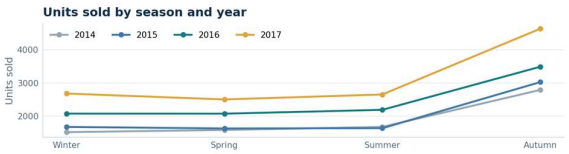


Fig. 4. Units sold by season and year

07 Key findings and business recommendations

- **Discount performance:** Profit margins were positive for discount bands up to 20% and negative for bands above 20%; the above-40% band showed a -77.40% margin. Review the underlying products and test stricter discount rules rather than assuming the discount alone caused each loss.
- **Seasonal demand:** Autumn recorded the highest units sold in each year shown (2014–2017). Use this as a planning signal to investigate, and validate against inventory availability and future-period data before adjusting stock.
- **Subcategory losses:** Tables recorded negative profit in all four season groups in the SQL output. Review item-level discounts, pricing and sales mix before deciding whether to adjust promotion or procurement.
- **Geography and shipping:** Investigate Central-region profitability and loss-making states such as Texas and Ohio. Standard Class produced the most total profit due to higher order volume, while First Class had the highest margin (13.93%); this is not evidence of lower shipping cost.

08 Limitations and data quality

The dataset contains only United States transactions, so country-to-country comparison is not possible. It contains no stock-on-hand, inventory-day, inventory-cost, production-capacity or shipping-cost fields; inventory turnover, exact production quantities and shipping-cost savings cannot be inferred. The cleaned file also contains a second “ship Date” column, and the tested Ship Date values match Order Date, producing zero-day delivery times. Delivery-speed conclusions are therefore excluded pending correction against the original source.

09 Conclusion

The analysis establishes a 12.47% overall profit margin and identifies meaningful differences by category, region, discount level and season. The most useful next actions are to investigate Furniture and Tables profitability, review heavily discounted products, and validate seasonal demand patterns against inventory data. The Tableau dashboard and SQL query file provide the evidence trail for these findings.

Submission files: Tableau dashboard (.twbx), SQL queries (.sql), Python notebook/script and generated chart outputs, this report (.pdf), and genuine evidence screenshots. Six Python charts and supporting summary CSVs were generated; retain those outputs with the notebook/script as evidence.